Disadvantage:
- not larger datasets
- overlapping classes

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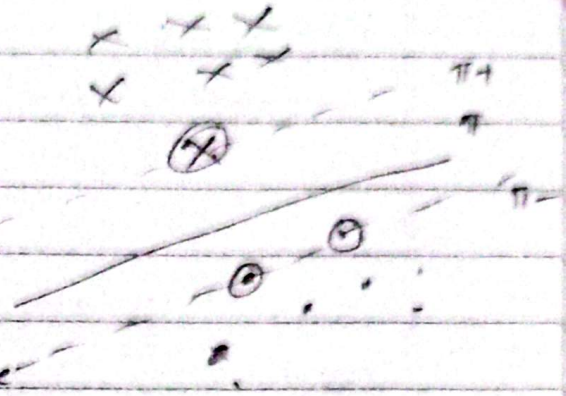
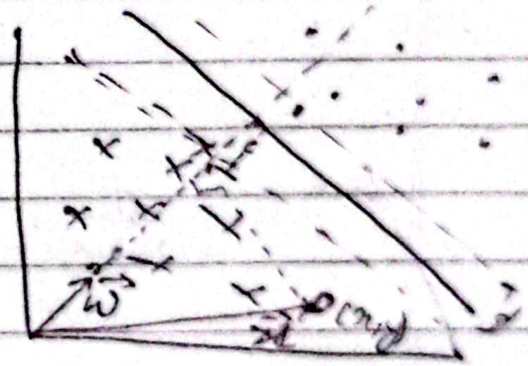
- Stop after getting 3rd positive & negative points [called support vectors] ⊗ & two ⊙ are support vectors.



- hyperplane should be at most distance from support vectors.

Advantages: stronger
- SVM are robust to outliers.
- Non-Linear data can also be solved by SVM.
- Both classification & Regression.
- Work with high dimensional data.

Mathematics of SVM. [Hard margin SVM] (we need perfect data without any error)



- Here, $\vec{w}$ is the vector
  - direction: perpendicular to 3 [π, π+, π-]
  - magnitude: unknown
- $\vec{u}$ (x, y) unknown vector whose classification is to be done.
- Take the projection of $\vec{u}$ in $\vec{w}$

$$\vec{w} \cdot \vec{u} \geq c$$

or; $\vec{w} \cdot \vec{u} - c \geq 0$

or; $\vec{w}\,\vec{u} + b \geq 0$

If true then +ve

If false the -ve point.

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Let 2-D:

eq^n of line: $\underset{w_1}{2}x + \underset{w_2}{3}y + \underset{b}{3} = 0$.

Let we have to find of point (3,2),
then;

$\vec{w} = (2,3)$ $\quad$ $\vec{x} = (3,2)$

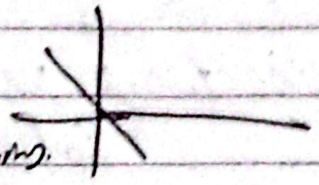
$w^T x$

\# For any $\vec{x}_i$

$$\hat{y} = \begin{cases} +1 & \text{if } \vec{w}.\vec{x}_i + b \geq 0. \\ -1 & \text{if } \vec{w}.\vec{x}_i - b < 0. \end{cases}$$

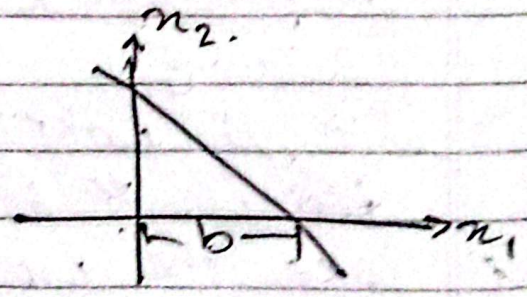
Q. $\underset{\text{label}}{y} = w^T x$

[if pass through origin]



$y = w^T x + b$

distance from origin to x-intercept.

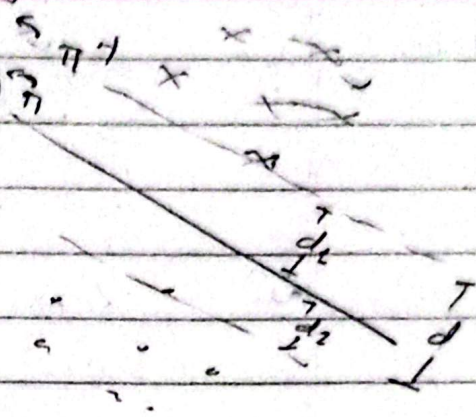


$\boxed{w^T x + b = 1}$

$\boxed{w^T x + b = 0}$ π^+

$d_1 = d_2$

$\boxed{w^T x + b = -1}$ π^-



- Why $w^T x + b = 1$
  Why (1) ??

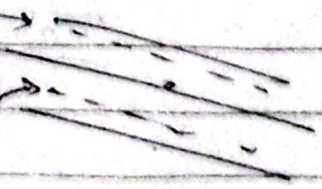
⇒ If 1, we get : x^2
If 25 & -25, we get : $25x^2$
If 37 & -37, we get : $37x^2$

- That above are the family of the line.

$2x + 3y + 3 = 0$
$2x + 3y + 3 = 1$ same
$2x + 3y + 3 = -1$

$\boxed{0.1 \times}$ $0.2x + 0.3y + 0.3 = 0$
$0.2x + 0.3y + 0.3 = 1$
$0.2x + 0.3y + 0.3 = -1$

$\boxed{10 \times}$ $20x + 30y + 30 = 0$
$20x + 30y + 30 = 1$
$20x + 30y + 30 = -1$



for $\times$, $\vec{w}\cdot\vec{x} + b \geq 1$.
For $\cdot$, $\vec{w}\cdot\vec{x} + b \leq -1$ } — (i)

Let label of +ve: $y_+ = +1$ve (y_i)
label of -ve: $y_- = -1$ve $(-y_i)$

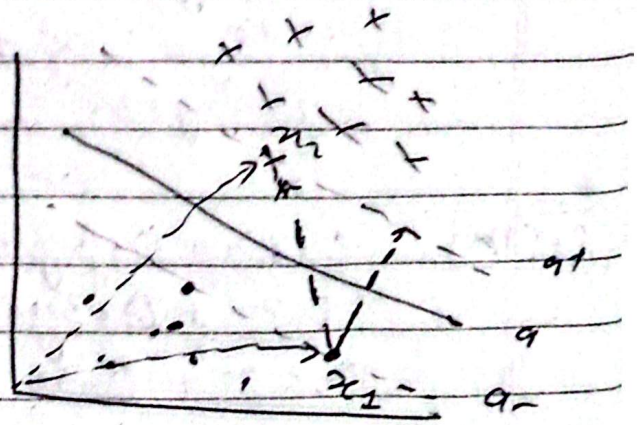
then eqn (i) will be:

$$y_i(\vec{w}\cdot\vec{x_i} + b) \geq 1. \quad \text{— (ii)}$$

For support vectors:

$$y_i(\vec{w}\cdot\vec{x_i} + \vec{b}) = 1$$

$\vec{w}$ is $\perp^r$ a, a_+, a_-



unit vector of w: $\hat{w} = \dfrac{\vec{w}}{\|w\|}$

$$d = (\vec{x_2} - \vec{x_1}) \cdot \frac{\vec{w}}{\|w\|}$$

$$= \frac{\vec{x_2}\cdot\vec{w} - \vec{x_1}\cdot\vec{w}}{\|w\|} \quad \text{— (i)}$$

Eqn of support vectors:

$$y_i(\vec{w}\cdot\vec{x} + \vec{b}) = 1.$$

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For +ve region: $y_i = 1$. $[x_i = x_2]$

$\vec{w}\vec{x_2} + \vec{B} = 1$.

$\Rightarrow \vec{w}\vec{x_2} = 1 - b$. — (ii)

For -ve region

$y_i = -1$. $[x_i = x_1]$

$-1(\vec{w}\vec{x_1} + \vec{b}) = 1$.

$\Rightarrow \vec{w}\vec{x_1} = -1 - b$ — (iii)

From (i) & (ii) & (iii):

$$d = \frac{(1-b) - (-1-b)}{\|w\|}$$

or, $$d = \frac{1 - b + 1 + b}{\|w\|}$$

$$\boxed{\therefore\ d = \frac{2}{\|w\|}}$$

If we take:

$$w^T x + b = \underset{\sim}{a} \longrightarrow d = \frac{2}{\|w\|} \times \underset{\sim}{a}$$

$$\left\{ \underset{(w^*, b^*)}{\operatorname{argmax}} \quad d = \frac{2}{\|w\|} \quad \text{such that} \quad y_i(\vec{w_i} \cdot \vec{x_i} + b) \geq 1 \right\}$$